

Study Questions: Systems

1. Define **Systems** in your own words, specifically as it applies to High School.
2. How does the Free Energy Principle constrain our understanding of Systems?
3. Contrast the Classical view of Systems with the Active Inference view.
4. Explain how a system of linear equations in algebra acts like a Markov blanket by constraining the possible values of variables. What role does the solution set play?
5. In the logistic growth equation $P(t+1) = rP(t)(1 - P(t)/K)$, identify which parts correspond to the agent's internal states, the environmental constraint, and the equilibrium the system seeks.
6. What does it mean mathematically for a system to "minimize free energy"? Describe this concept using an example from algebra or calculus that you have encountered in class.
7. How does adding a new equation to a system of equations parallel the arrival of new sensory evidence in Active Inference? What happens to the uncertainty in both cases?
8. Compare and contrast a closed system and an open system in thermodynamics. Which type better represents an Active Inference system, and why?
9. Draw a diagram showing how the perception-action loop operates within a mathematical system. Label the prediction, observation, and prediction error components.
10. If a system is at equilibrium, does that mean it has zero free energy? Explain your reasoning using the logistic growth model as a reference.
11. Describe how feedback loops in dynamical systems relate to the concept of active inference. Give a mathematical example of positive and negative feedback.

12. How would you use a state-space representation (vectors and matrices) to model a simple Active Inference system? What would the state vector contain?
13. Explain why the carrying capacity K in the logistic equation acts as a boundary condition. How is this analogous to a Markov blanket in Active Inference?
14. A student claims that a random number generator is a "system" in the Active Inference sense. Evaluate this claim -- what properties would be missing?
15. How does the concept of convergence in a mathematical sequence relate to a system reaching a stable equilibrium in Active Inference?
16. Design a simple system of two equations with two unknowns. Walk through how solving the system step-by-step mirrors the process of an agent reducing uncertainty about its environment.
17. What mathematical tools (e.g., eigenvalues, derivatives, probability distributions) would you need to analyze whether a system is stable or unstable? Connect each tool to an Active Inference concept.
18. Explain the difference between a system's parameters and its states. In the logistic growth model, which quantities are parameters and which are states?
19. How might you represent the "surprise" experienced by a mathematical system when it encounters data that conflicts with its current model? What formula would quantify this?
20. Synthesize the ideas from this module: propose a novel mathematical system (not one from the examples) that could be analyzed using Active Inference principles. Identify its boundary, internal states, and the quantity it minimizes.